FlavourBench: Executable Culinary Evaluation of Frontier Language Models

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Abstract

We introduce FlavourBench, an executable benchmark for culinary reasoning in frontier language models. Epicure, a versioned culinary system, computes the answer to each task before any evaluated model is called. No human labels or model judge enter the score. We test 20 frontier endpoints, including GPT-5.6, Claude 5, Gemini 3, Kimi K3, Qwen 3.8 Max, GLM 5.2, DeepSeek V4, and Cohere Command A, on the same 32 four-choice tasks across substitution, composition, cookability, and cultural evidence. Model-only accuracy defines the FlavourBench Score. GPT-5.6 Sol Pro scores 62.5% (20/32); Claude Sonnet 5 and DeepSeek V4 Pro each score 59.4% (19/32). Exact paired tests do not separate these three on the current panel ($p = 1.000$). A second, named-operation condition tests whether models can call Epicure and return its result. 14 of 19 responding endpoints reach 100%, so this condition is reported as an execution diagnostic rather than a ranking metric. The public release contains 1280 model-task-condition records with routes, completion states, traces, and content hashes. A dependency-free verifier recomputes every score, table, and figure offline.

1 Introduction

Benchmarks are easiest to trust when their ground truth is external to the models being tested. Code can be executed, database states can be inspected, and games can be scored. Culinary reasoning rarely has that luxury. Ingredient substitutions, pairings, technique choices, and cuisine associations are commonly assessed by another language model or by a small set of human preferences. The evaluator then becomes part of the construct.

FlavourBench starts from Epicure, a read-only culinary runtime with 1,790 ingredients represented in a 300-dimensional space [20, 21]. Four runtime operations are converted into four-choice questions. The compiler records each answer before it contacts an evaluated model.

The design asks one ranking question and one diagnostic question. The FlavourBench Score measures how often a model endpoint returns Epicure's answer without runtime access. The same endpoint then receives the task with one read-only Epicure call, providing a controlled execution check. Every model receives every task in both conditions. Only the model-only score determines rank.

Our release evaluates 20 current endpoints, including GPT-5.6 Sol, Terra, and Luna; Claude Fable, Opus, and Sonnet 5; Gemini 3.1 Pro and 3.6 Flash; Grok 4.5; Kimi K3; Qwen 3.8 Max; GLM 5.2; two DeepSeek V4 variants; MiniMax M3; Nemotron 3 Ultra; Mistral Large 2512; Tencent HY 3; and Cohere Command A Plus and Command A Reasoning. Kimi and both Cohere models use direct provider routes. Every other endpoint uses an exact, no-fallback OpenRouter route selected before the run.

The paper makes three contributions:

1. an automated culinary benchmark whose questions and answer keys come from a versioned executable system rather than a model judge;
2. a common-task frontier evaluation with 640 matched pairs, exact missing-response penalties, and a public 20-row leaderboard;
3. a matched, named-operation diagnostic that exposes tool-integration failures and task-family interactions without changing the ranking score.

Epicure has two roles. It fixes the answer key, then becomes the intervention in the second arm. This keeps the leaderboard and the Epicure comparison connected without combining them into an arbitrary composite score.

2 Epicure reference answers

2.1 A versioned reference

For FlavourBench, ground truth is the output of a specific version of Epicure. It is not a claim about universal culinary preference. The release binds the application, evidence bundle, release identifier, ingredient count, and embedding dimensionality by hash. A different runtime produces a different task-set hash.

This definition makes every answer independently recomputable. It also gives the score a direct interpretation: accuracy is agreement with the published culinary runtime under the published task construction. No second model decides which response sounds better.

2.2 Four balanced task families

The compiler creates eight tasks in each of four families. Seeds and ingredient pairs are fixed before evaluation. For every task, Epicure is called once to obtain the

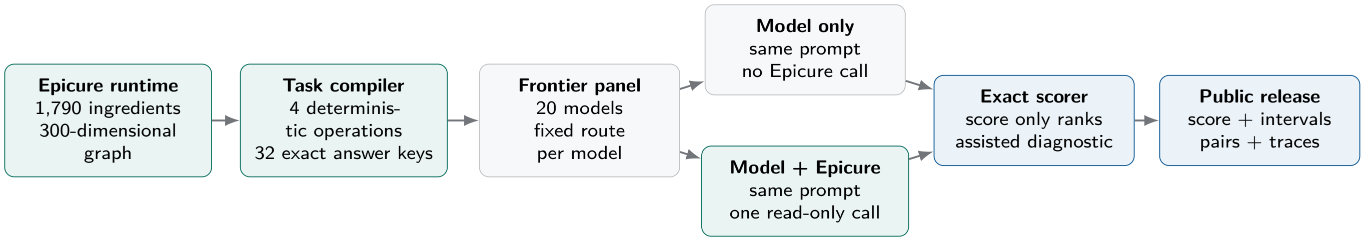

Figure 1: The FlavourBench pipeline. Epicure fixes each answer before evaluation. The Model only arm produces the ranked FlavourBench Score. The matched Model + Epicure arm tests execution with the answer-generating system available and does not affect rank. Both reconstruct from the same public release.

reference result. The correct answer is extracted programmatically, three distractors are constructed without model generation, and the four options are shuffled with a task-ID-derived seed.

The task families deliberately mix retrieval and computation. Neighbour and cuisine questions test whether a model knows relationships encoded by the graph. Pairing questions require an exact numeric result that is difficult to guess. Axis questions test a comparative claim rather than a memorised scalar. Together they prevent the leaderboard from reducing to one operation or one kind of culinary trivia.

2.3 Prompt and answer contract

Each prompt names the relevant operation and gives four choices. Models may explain briefly, but the last answer line must be `FINAL_CHOICE: X`. The parser accepts A, B, C, or D on that marker line. A missing marker, abnormal provider finish, timeout, or absent response scores zero. The rule fits in the standalone verifier shipped with the release.

The complete task set contains no hidden human label and no model-authored item. The benchmark is therefore suitable for automatic reruns, CI-style regression tests, and training loops where thousands of candidate responses must be scored consistently.

3 Evaluation design

3.1 One score and one matched intervention

For model m and task t, let M_{mt} equal one when the Model only response finishes normally and matches the Epicure answer. Let E_{mt} be the same indicator for Model + Epicure. The prompt, route, temperature, top-p, seed request, output limit, and answer parser are identical. Only access to Epicure changes.

Each Model + Epicure response may make one read-only call to the operation named in the prompt. The runner records the operation, arguments, latency, error flag, and a hash of the returned payload. The returned data is bounded before it enters model context. Automatic provider fallback is disabled, so a route failure remains a failure for that route.

The single ranking metric is the FlavourBench Score:

$$F_m = 100\,\frac{1}{32}\sum_t M_{mt}. \tag{1}$$

Every family contains eight tasks, so ordinary accuracy already gives each family equal weight. Missing or invalid responses remain in the denominator as incorrect. The paper and public Space give equal values of F_m the same score rank; model name orders rows within a tie. No assisted metric breaks a FlavourBench Score tie.

Epicure Gain is the matched difference

$$G_m = 100\,\frac{1}{32}\sum_t \left(E_{mt} - M_{mt}\right), \tag{2}$$

reported in percentage points. G_m is not part of the FlavourBench Score. The release also keeps the four paired outcomes for every task: both correct, Model only correct, Model + Epicure correct, or neither correct.

3.2 Completion and uncertainty

Completion is the fraction of assigned responses that end normally with a parseable final marker. The score still counts missing responses as incorrect. Reporting completion beside the score keeps endpoint failure separate from a wrong culinary answer. A zero score with zero completions is not evidence of zero model capability. For each 32-task accuracy we report a 95% Wilson interval as a descriptive uncertainty indicator. The panel is deliberately balanced rather than randomly drawn from all culinary questions, so the interval is not a population guarantee.

The benchmark is designed for visible differences, not hundredths of a point. One answer changes the score by 3.125 points. Exact paired McNemar tests comparing the 20/32 leader with each 19/32 runner-up give $p = 1.000$ and $p = 1.000$. These unadjusted descriptive tests, the overlapping intervals, family breakdowns, and task matrix all point to the same reading: this release resolves broad separation, not the order of adjacent leaders.

Family	Epicure operation	Exact target	Example query	Tasks
Substitution	`neighbors`	first ranked neighbour	top neighbour for tomato	8
Composition	`pairing_score`	four-decimal affinity	tomato + basil	8
Cookability	`compare_on_axis`	larger flavour projection	coffee vs. chocolate on bitter	8
Evidence	`cultural_profile`	highest cuisine cosine	cuisine direction for kimchi	8

Table 1: The task set is balanced by construction. Each family maps one deterministic Epicure operation to a four-choice answer. Chance accuracy is 25%.

3.3 Twenty-model frontier panel

The panel was fixed before the final calls. It includes three GPT-5.6 tiers rather than treating one deployment as the entire OpenAI family; three Claude 5 tiers; two Gemini tiers; two DeepSeek V4 routes; both current Cohere Command A routes; and a broad set of current independent and open-weight families. Kimi K3 runs through Kimi's Anthropic-compatible endpoint. Command A Plus and Command A Reasoning run through Cohere V2 Chat. Qwen 3.8 Max and GLM 5.2 use explicit OpenRouter routes in the automated batch [12, 16, 3, 5, 4, 17]. The exact 20-row route table appears in Appendix A.

4 Results

4.1 The FlavourBench Score leaderboard

GPT-5.6 Sol Pro leads the fixed panel with 20/32 correct and a FlavourBench Score of 62.5 (Wilson 95% interval 45.3–77.1). Table 2 gives the score-only ranking, correct counts, intervals, and parsed-answer counts. The top three differ by one answer, their intervals overlap substantially, and the paired tests above do not separate the leader from either runner-up. Across all 19 leader comparisons, Holm adjustment separates only 3 rows, Grok 4.5, Mistral Large 2512, Nemotron 3 Ultra; two are DNF endpoints. Models with no parseable model-only answer are marked DNF rather than interpreted as the least capable systems.

The answer key exists before the model call, every endpoint receives the same questions, and no judge is sampled after the fact. Missing outputs count as incorrect in the operational score but remain visible in the "Parsed" column. Mistral Large 2512 returned 32 normal model-only responses, but none contained a parseable final choice. Nemotron 3 Ultra returned no response in either condition. Both zeroes are operational outcomes rather than capability estimates.

4.2 What Epicure changes

The assisted condition is deliberately secondary. Among the 19 endpoints that returned assisted completions, 14 score 100% and mean accuracy is 98.8%. FlavourBench Score and Epicure Gain correlate $r = -0.990$ because a near-100% assisted ceiling makes gain approximately $100 - F_m$. Gain therefore adds little ranking information and does not appear in the primary table.

The assisted prompt names the relevant operation, whose output is also used to compile the answer key. Near-ceiling assisted accuracy is therefore expected. The condition tests whether an endpoint can call the specified operation, read its output, and preserve the answer contract. It is not an independent validation of Epicure, a second capability score, or a test of open-ended tool discovery.

4.3 Families explain aggregate ties

The four task families have different demands. Pairing questions require an exact four-decimal value; substitution and cultural questions can sometimes be answered from broad food knowledge; axis comparisons require a directional decision in Epicure's latent space. Figure 4 shows that similar total scores can arise from different profiles.

4.4 Every model and task pair is inspectable

Aggregate bars can hide a benchmark bug or a single pathological task. Figure 5 therefore publishes all 640 paired outcomes. Blue cells are stable knowledge: both conditions are correct. Gold cells are Epicure wins: Model only is wrong and Model + Epicure is correct. Red cells show the reverse outcome. Grey cells show questions neither condition solved.

4.5 Cohere, Kimi, Qwen, and route fidelity

Both Cohere models are full members of the public panel. Command A Plus has score rank 16 with a FlavourBench Score of 40.6; Command A Reasoning has score rank 9 at 46.9. Kimi K3 has score rank 4 through the direct Kimi endpoint. Qwen 3.8 Max has score rank 9 through an exact OpenRouter route, and GLM 5.2 has score rank 14. No route changes during generation.

The two direct Cohere routes each produced all 64 assigned arms. Their returned identifiers are `command-a-plus-05-2026` and `command-a-reasoning-08-2025`, respectively; the direct Kimi route likewise produced all 64 arms under returned identifier `k3`. The offline verifier checks these provider, identity, and arm-count contracts before accepting the release.

This matters because a model name is not a transport

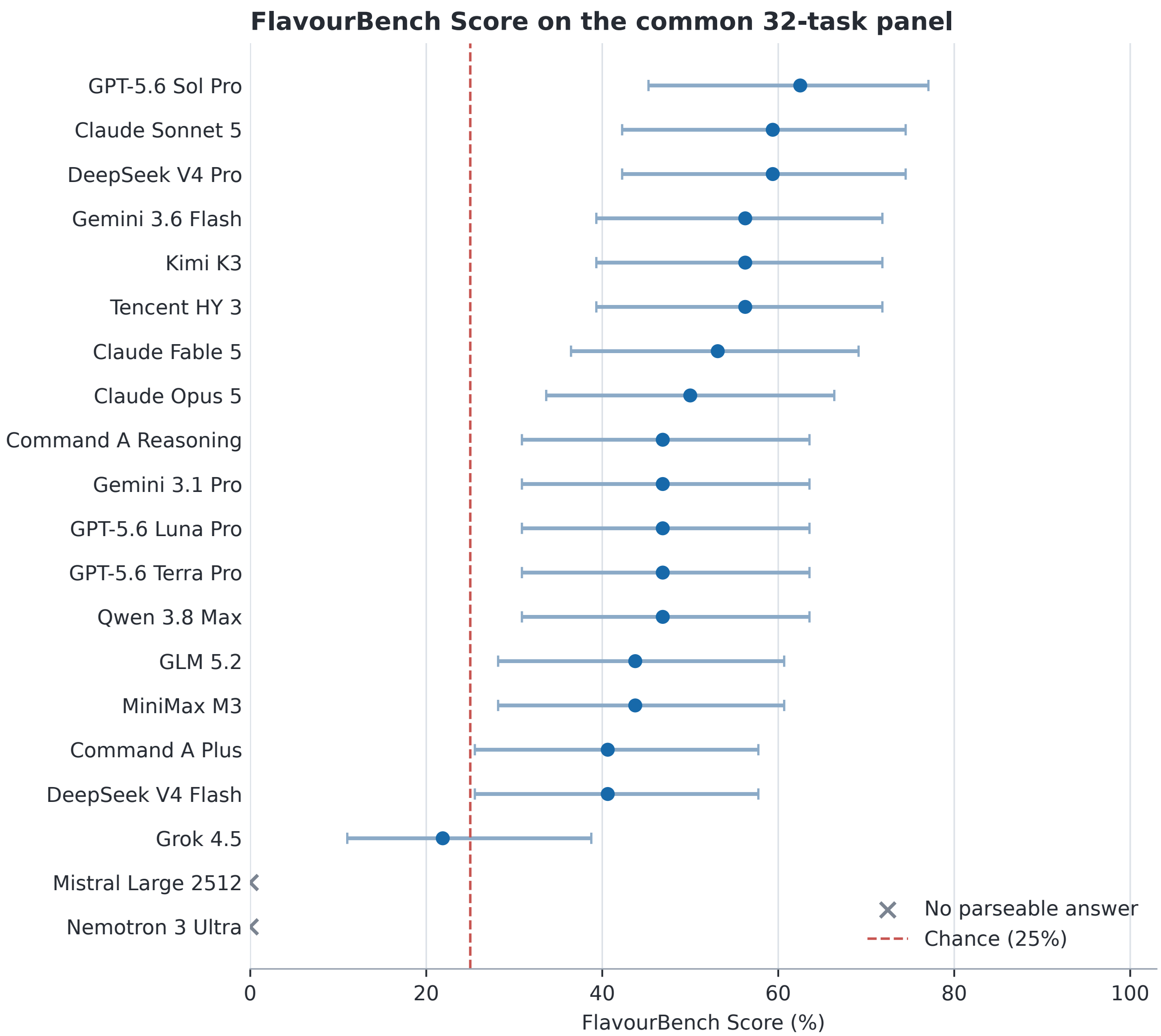

Figure 2: FlavourBench Score against Epicure answer keys. Points are exact accuracy over the common 32-task panel; bars are descriptive 95% Wilson intervals. Crosses mark endpoints with no parseable model-only answer. The dashed line is four-choice chance.

contract. The release stores the requested model, returned model, provider, route, generation ID, latency, token counts, and response hash. Direct and routed endpoints can therefore be analysed separately without pretending they are interchangeable deployments.

4.6 Score and response time

Figure 6 plots FlavourBench Score against median response time. Marker colour identifies the execution route. Accuracy and latency are separate operational dimensions: faster endpoints do not necessarily score higher, and direct-provider routes are not pooled with routed deployments.

5 Real prompts, answers, and tool calls

A benchmark paper should show what was actually asked. The four cases below are selected mechanically, one per family. Where possible, the builder chooses a model and task that is wrong in Model only and correct in Model + Epicure. Each case includes the full prompt, both final answers, the actual operation and arguments, and the exact answer key.

Case 1: Substitution with GPT-5.6 Sol Pro. **Prompt.** FlavourBench Epicure-native exact task. Use the read-only culinary evidence tools if they are available; otherwise answer from your own knowledge.

According to Epicure's 'neighbors' result for 'tomato' with top_k=4, which ingredient is ranked first?

Choices: A. oregano B. bell_pepper C. olive_oil D.

Score rank	Model	Correct	FB Score	Wilson 95%	Parsed
1	GPT-5.6 Sol Pro	20/32	62.5	45.3 to 77.1	31/32
2	Claude Sonnet 5	19/32	59.4	42.3 to 74.5	32/32
2	DeepSeek V4 Pro	19/32	59.4	42.3 to 74.5	32/32
4	Gemini 3.6 Flash	18/32	56.2	39.3 to 71.8	32/32
4	Kimi K3	18/32	56.2	39.3 to 71.8	32/32
4	Tencent HY 3	18/32	56.2	39.3 to 71.8	32/32
7	Claude Fable 5	17/32	53.1	36.4 to 69.1	30/32
8	Claude Opus 5	16/32	50.0	33.6 to 66.4	32/32
9	Command A Reasoning	15/32	46.9	30.9 to 63.6	30/32
9	Gemini 3.1 Pro	15/32	46.9	30.9 to 63.6	30/32
9	GPT-5.6 Luna Pro	15/32	46.9	30.9 to 63.6	32/32
9	GPT-5.6 Terra Pro	15/32	46.9	30.9 to 63.6	32/32
9	Qwen 3.8 Max	15/32	46.9	30.9 to 63.6	28/32
14	GLM 5.2	14/32	43.8	28.2 to 60.7	32/32
14	MiniMax M3	14/32	43.8	28.2 to 60.7	31/32
16	Command A Plus	13/32	40.6	25.5 to 57.7	32/32
16	DeepSeek V4 Flash	13/32	40.6	25.5 to 57.7	24/32
18	Grok 4.5	7/32	21.9	11.0 to 38.8	10/32
DNF	Mistral Large 2512	0/32	0.0	0.0 to 10.7	0/32
DNF	Nemotron 3 Ultra	0/32	0.0	0.0 to 10.7	0/32

Table 2: The score-only public leaderboard. “Correct” and “FB Score” use model-only answers; equal scores share a score rank. Wilson 95% is a descriptive interval for the fixed 32-task panel, and “Parsed” reports responses with a valid final choice. DNF means that the endpoint returned no parseable model-only answer.

Statistical indicator	Observed value	Reading
Score resolution	$1/32 = 3.125$ points	Adjacent rows often differ by one answer.
Leader uncertainty	62.5%; Wilson 95% 45.3–77.1%	Exact for this panel; broad descriptive interval beyond it.
Leader vs. 19/32 runners-up	Exact paired $p = 1.000$ and $p = 1.000$	The fixed panel does not separate the top three.
Leader vs. full panel	3/19 comparisons significant after Holm adjustment	Only broad operational separation is resolved.
Assisted ceiling	14/19 endpoints at 100%	Useful for integration failures, not rank discrimination.
Score–gain correlation	$r = -0.990$	Near-perfect inverse dependence; gain is largely the distance to the assisted ceiling.

Table 3: The statistical reading of this release. Intervals and tests are descriptive because the 32 tasks form a fixed balanced panel rather than a random sample of all culinary questions.

Family	Epicure operation	Tasks	Model only	Model + Epicure
Substitution	`neighbors`	8	18.1	94.4
Composition	`pairing_score`	8	40.6	94.4
Cookability	`compare_on_axis`	8	62.5	92.5
Evidence	`cultural_profile`	8	54.4	94.4

Table 4: Panel-average performance by family and the corresponding reference operation.

onion

In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE: X‘, replacing X with A, B, C, or D.

Model only. `FINAL_CHOICE: D` **Model + Epicure.** `FINAL_CHOICE: B`

Epicure call. `neighbors({"ingredient": "tomato", "top_k": 4})`. The answer key is `FINAL_CHOICE: B`.

Epicure result. `{"ingredient": "tomato", "neighbors": [{"name": "bell_pepper", "rank": 1, "sim": 0.4799}, {"name": "olive_oil", "rank": 2, "sim": 0.4786}, {"name": "oregano", "rank": 3, "sim": 0.4646}, {"name": "onion", "rank": 4, "sim": 0.4567}]}`

Case 2: Composition with DeepSeek V4 Pro. **Prompt.** FlavourBench Epicure-native exact task. Use the read-only culinary evidence tools if they are available; otherwise answer from your own knowledge.

What exact four-decimal ‘pairing_score‘ does Epicure return for ‘tomato‘ and ‘basil‘?

Choices: A. 0.2707 B. 0.2407 C. 0.3207 D. 0.1907

In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE: X‘, replacing X with A, B, C, or D.

Model only. `FINAL_CHOICE: B` **Model + Epicure.** `FINAL_CHOICE: A`

Epicure call. `pairing_score({"ingredient_a": "tomato", "ingredient_b": "basil"})`. The answer key is `FINAL_CHOICE: A`.

Epicure result. `{"all_pairs_range": {"median": 0.092, "p10": 0.0197, "p25": 0.0521, "p75": 0.1392, "p90": 0.1899}, "pairing_score": 0.2707, "percentile_label": "very high (>=p90)", "resolved_a": "tomato", "resolved_b": "basil"}`

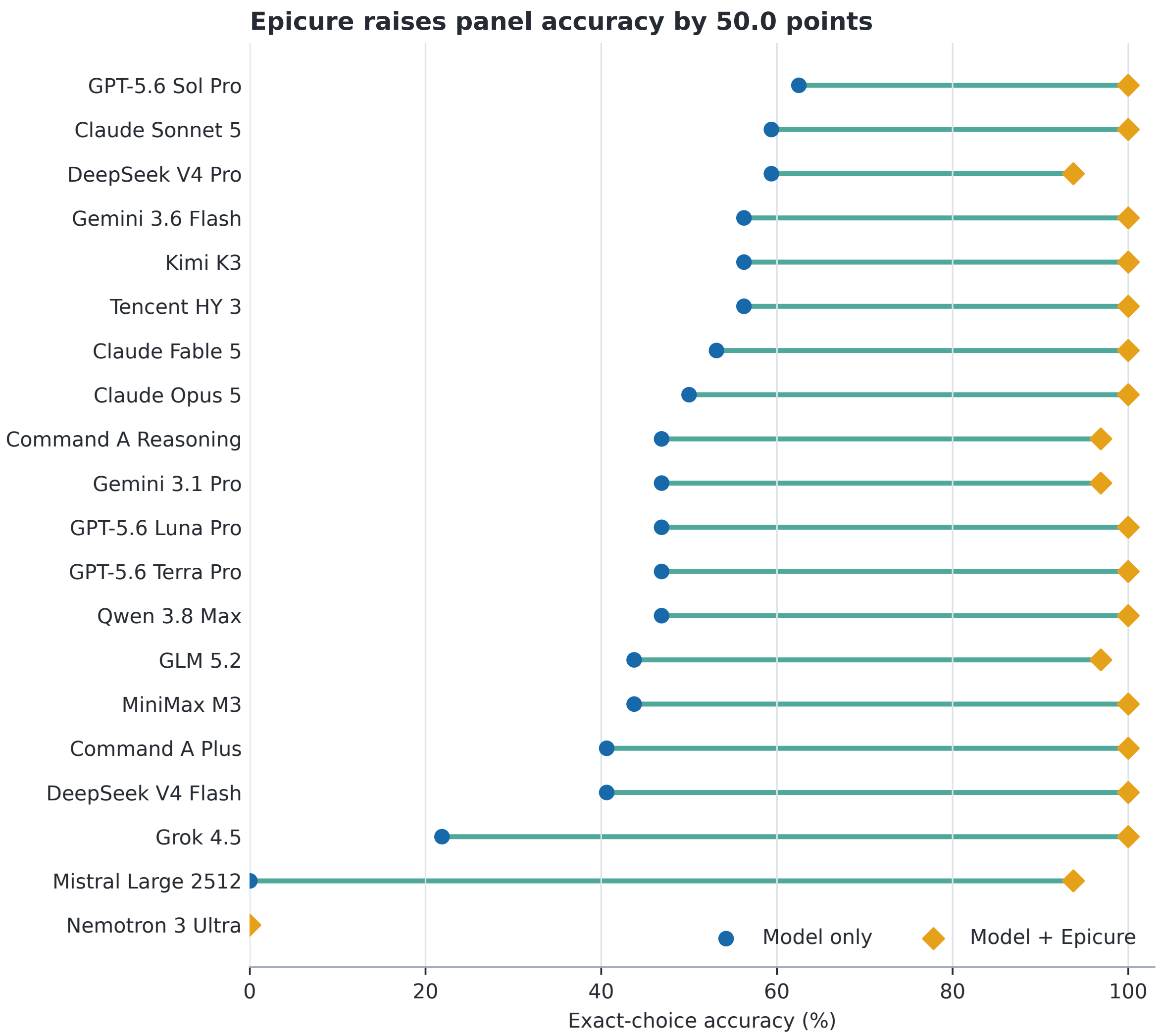

Figure 3: Model-only and named-operation assisted accuracy over the same 32 tasks. This secondary diagnostic shows the assisted ceiling and endpoint failures; it does not determine rank.

Case 3: Cookability with GPT-5.6 Sol Pro. Prompt. FlavourBench Epicure-native exact task. Use the read-only culinary evidence tools if they are available; otherwise answer from your own knowledge.

Which statement matches Epicure's ‘compare_on_axis‘ result for ‘coffee‘ versus ‘chocolate‘ on ‘cf_bitter‘?

Choices: A. coffee has the higher projection B. the projections are equal to four decimals C. Epicure cannot resolve one or both ingredients D. chocolate has the higher projection

In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE: X‘, replacing X with A, B, C, or D.

Model only. `FINAL_CHOICE: A` **Model + Epicure.** `FINAL_CHOICE: D`

Epicure call. `compare_on_axis({"axis": "cf_bitter", "ingredient_a": "coffee", "ingredient_b": "chocolate"})`. The answer key is `FINAL_CHOICE: D`.

Epicure result. `{"axis": "cf_bitter", "axis_range": {"p10": -0.0877, "p90": 0.1429}, "delta": 0.0835, "label_a": "moderate", "label_b": "high", "projection_a": 0.0076, "projection_b": 0.0911, "resolved_a": "coffee", "resolved_b": "chocolate"}`

Case 4: Evidence with Grok 4.5. Prompt. FlavourBench Epicure-native exact task. Use the read-only culinary evidence tools if they are available; otherwise answer from your own knowledge.

Which cuisine direction has the highest cosine score in Epicure's ‘cultural_profile‘ result for ‘miso‘?

Choices: A. Japanese B. East_Asian C. Southeast_Asian D. South_Asian

In your final answer, explain briefly if useful and finish on its own line with exactly ‘FINAL_CHOICE: X‘, replacing X with A, B, C, or D.

Model only. `{"name": "cultural_profile", "arguments": {"ingredient": "miso"}}` **Model**

+ **Epicure.** `FINAL_CHOICE: A`

Epicure call. `cultural_profile({"ingredient": "miso"})`. The answer key is `FINAL_CHOICE: A`.

Epicure result. `{"cuisines": {"East_Asian": {"range": {"p10": -0.1649, "p90": 0.3039}, "score": 0.1625}, "Eastern_European": {"range": {"p10": -0.1466, "p90": 0.1034}, "score": -0.1494}, "Japanese": {"range": {"p10": -0.1359, "p90": 0.1896}, "score": 0.3227}, "Latin_American": {"range": {"p10": -0.1672, "p90": 0.1281}, "score": -0.1538}, "Mediterranean": {"range": {"p10": -0.2323, "p90": 0.206}, "score": -0.1198}, "South_Asian": {"range": {"p10": -0.0808, "p90": 0.071}, "score": -0.0585}, "Southeast_Asian": {"range": {"p10": -0.1625, "p90": 0.193}, "score": 0.0426}, "Western_Atlantic": {"range": {"p10": -0.2304, "p90": 0.183}, "score": -0.1546}}, "note": "Cosine similarity to each cuisine direction. Use p10/p90 to judge whether a score is typical or extreme.", "resolved": "miso"}`

These examples show exactly what the two measurements contain. Model only tests knowledge or inference about the hidden graph. Model + Epicure names the operation, so it tests whether the model can form valid arguments, read the returned structure, and preserve the answer contract. A correct choice with the wrong operation still earns answer credit, but the release separately reports whether the reference operation was matched. Across the run there are 606 projected EPICURE calls and 606 model and task pairs matching the reference operation.

6 Using the score as a training signal

The same environment exposes four binary indicators for a generated trajectory τ: r_{answer} for a correct answer, r_{format} for a parseable final marker, r_{call} for a successful Epicure call, and r_{align} for a call matching the reference operation. The public leaderboard uses r_{answer} in Model only. A future training objective could use

$$r(\tau) = r_{\text{answer}} r_{\text{format}} + \lambda r_{\text{answer}} r_{\text{call}} + \mu r_{\text{align}}, \tag{3}$$

where each component is a binary indicator and λ, μ set the value of successful and reference-matching calls. Requiring a correct answer prevents irrelevant calls from earning the main reward. The interface could support rejection sampling, preference optimisation, or reinforcement learning on new ingredient seeds. This paper does not train a checkpoint; such a study should keep the public leaderboard as a final evaluation split.

7 Reproducibility and release

The public release is one JSON document with hash `197b84d75cca...`. It contains the 20 route records, 32 full prompts, answer keys and reference EPICURE results, 1,280 observation rows, final answer markers, response times, and bounded Epicure-trace projections. 1195 provider responses were observed; 85 missing arms remain explicit zero-scored rows.

The leaderboard can be reconstructed without a network connection, provider key, GPU, or running EPICURE service:

```
python3 -I reproduce_epicure_native.py \
  --release generated/epicure-native/\
  epicure-native-release.json
```

The verifier rejects duplicate JSON keys, non-finite numbers, a changed content address, an incomplete $20 \times 32 \times 2$ grid, or any leaderboard value that differs from recomputation. The paper tables and figures are generated from the same release, so a changed score cannot leave a stale chart behind.

Reproducing the original hosted calls is a different operation from replaying the result. The release records exact routes and request policies, but provider deployments can change. A fresh frontier run should generate a new manifest and release rather than overwrite this snapshot.

8 Discussion

What the score means. The FlavourBench Score answers a narrow operational question: how often did this deployed endpoint return the answer fixed by the published EPICURE runtime? It includes endpoint availability because missing responses score zero. The completion columns reveal that component. The score is not a universal measure of cooking skill, but it is exact, reproducible, and independent of a model judge.

Do we need humans? Not for this leaderboard. Humans are useful when the target is preference, creativity, safety, or the quality of an open-ended recipe. They are unnecessary when the question is whether a model's choice matches a deterministic runtime result. FLAVOURBENCH uses that distinction to automate the core ranking while leaving subjective culinary evaluation as a separate track.

Why run Model + Epicure? The condition is a named-operation execution test. Because EPICURE both generated the answer key and supplies the relevant runtime output, its near-ceiling accuracy is expected. The condition still catches tool-call, argument, parsing, and answer-contract failures, but it neither ranks models nor independently validates EPICURE. The dumbbell and pair matrix expose those failures task by task.

Scaling the benchmark. The task compiler can expand the seed lists while preserving the four operations. The next release should reserve unseen ingredients and pairs, remove near-duplicate questions, and measure stability across runtime versions. As a simple scale guide near 50% accuracy, 100 independent tasks give about a 10-point 95% half-width and 400 give about five points; paired rank claims should be powered on the observed task covariance. The current panel is a frontier snapshot and a reproducible template for that larger hidden set.

9 Related work

HELM argues for explicit scenarios and metrics, while recent benchmark audits emphasise validity, contamination, and transparent reporting [13, 23]. Chatbot Arena and WildBench evaluate broad answer quality through preferences or checklists [1, 14]. FlavourBench instead targets an executable domain model and needs no judge for its primary score.

ToolLLM, BFCL, and τ-bench make tool invocation or environment state observable [19, 18, 28]. FlavourBench adds a matched Model only condition, so Epicure becomes an intervention rather than a requirement shared by every response. LiveBench, LiveCodeBench, GPQA, MMLU-Pro, and SWE-bench illustrate complementary strategies for reducing contamination or grounding answers in difficult tasks and executable outcomes [26, 9, 22, 25, 10].

Culinary datasets such as FlavorDB and Recipe1M organise ingredient and recipe evidence [7, 24]. Recent food benchmarks emphasise other constructs: NutriBench estimates carbohydrates from meal descriptions; FoodBench-QA grounds nutrition tasks in databases and ontologies; DiningBench tests multimodal dish perception and nutrition reasoning; and FAM-Bench evaluates condition-aware food-as-medicine decisions [8, 6, 11, 15]. CookingSense and planning-oriented culinary work test commonsense and procedures [2, 27]. FlavourBench targets a complementary construct: ingredient-space flavour decisions whose targets are compiled from a callable, versioned runtime, with access to that runtime measured as a matched intervention.

10 Limitations

The 32-task panel resolves broad differences, not adjacent leaders: one answer moves the score by 3.125 points and the leading intervals overlap. The English ingredient seeds cover four Epicure operations, hosted routes are a dated snapshot, and public prompts may eventually be memorised. Agreement with Epicure inherits the runtime's representation choices; it is not a food-safety or cultural-authenticity certification. A larger hidden panel, additional operations, and repeated route snapshots are the immediate next tests.

11 Conclusion

FlavourBench turns a versioned culinary runtime into an executable benchmark for frontier language models. GPT-5.6 Sol Pro leads this 20-model, 32-task snapshot at 62.5%, while the overlapping leading intervals make the close group explicit. Scores, completions, prompts, responses, and traces all reconstruct from the released 1,280-arm grid. The result is a simple automated leaderboard with a separate named-operation diagnostic, not a model judge hidden behind another score.

Benchmark	Scope	Primary target	Paired tool?
Chatbot Arena	general chat	pairwise human preference	No
WildBench	general chat	task-specific checklist	No
BFCL	function calling	executable call match	No
τ-bench	interactive agents	exact goal state	No
DiningBench	multimodal food	dish and nutrition labels	No
FAM-Bench	food-as-medicine	expert suitability label	No
FlavourBench	culinary reasoning	versioned executable Epicure operation	Yes

Table 5: FlavourBench complements broad, tool-use, and food-domain benchmarks. Its distinctive unit is the same endpoint and task measured as Model only and Model + Epicure; "paired tool" denotes that matched intervention rather than ordinary tool availability.

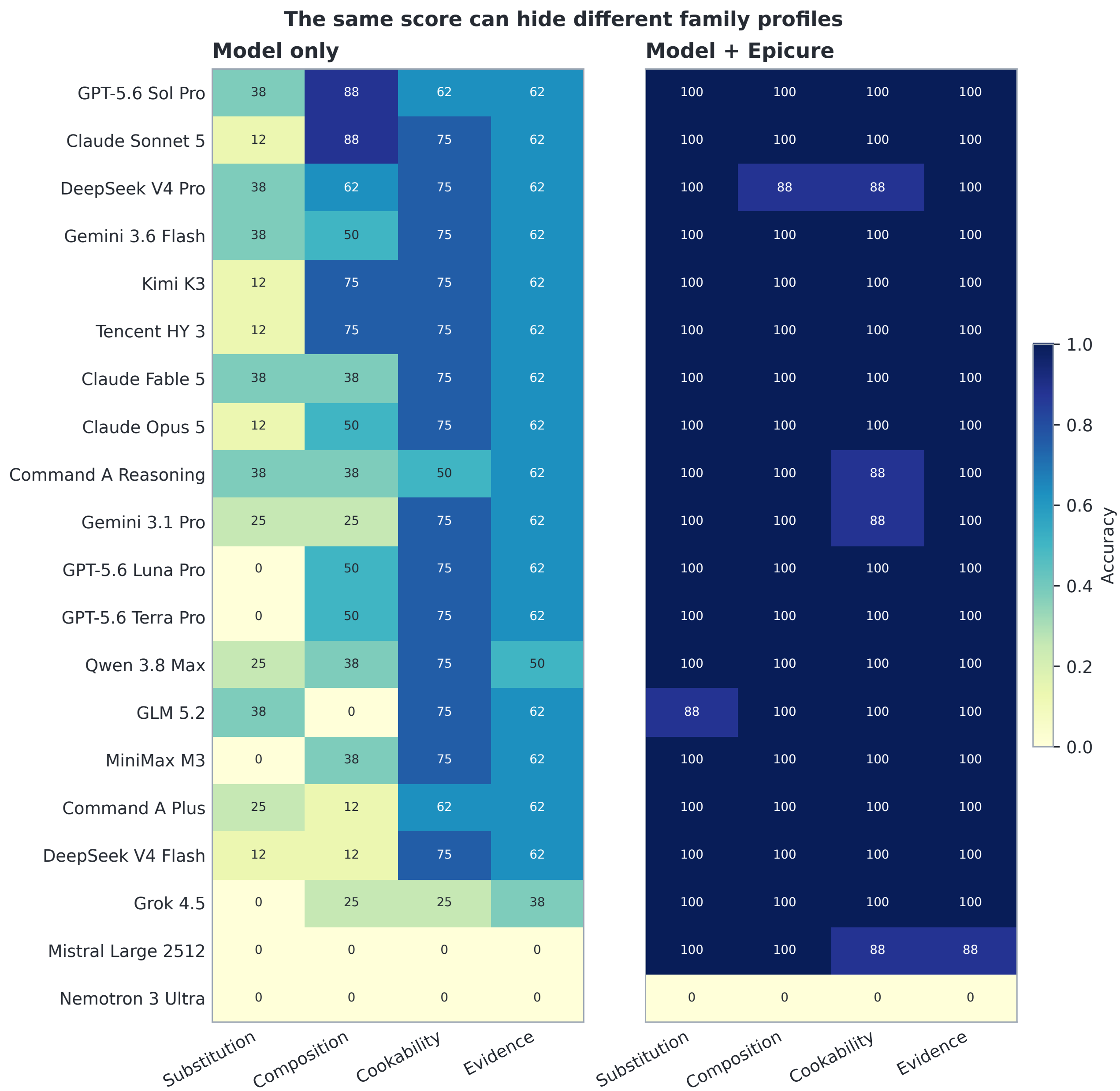

Figure 4: Model only and Model + Epicure accuracy by task family. Each value is a percentage over eight tasks. Similar FlavourBench Scores can come from different family profiles.

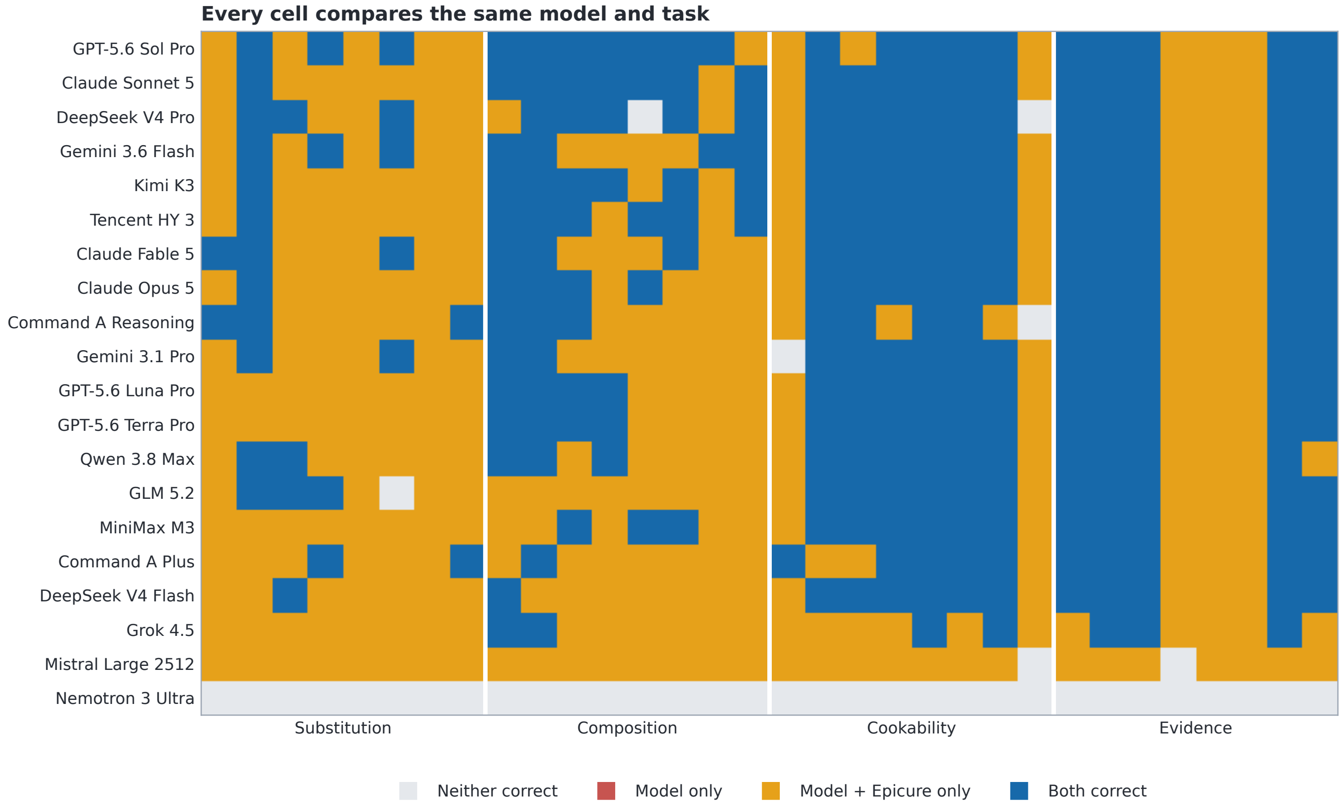

Figure 5: All 640 matched outcomes, ordered by FlavourBench Score and grouped by task family. The matrix shows which exact tasks produce Epicure Gain and where the assisted response regresses.

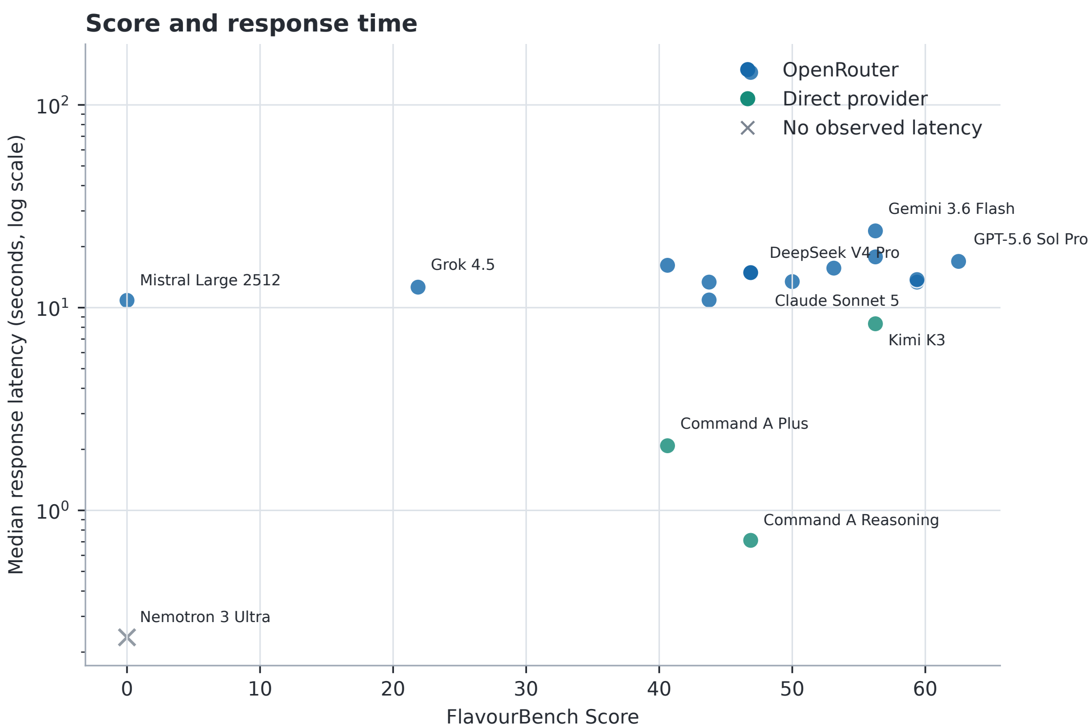

Figure 6: FlavourBench Score and response time. Response time uses a log scale; colour identifies direct-provider and OpenRouter execution. An × marks a route with no observed response time.

A Exact execution routes

Slot	Model	Execution route
1	GPT-5.6 Sol Pro	openrouter: openai/flex
6	Claude Sonnet 5	openrouter: amazon-bedrock/claude-on-aws
13	DeepSeek V4 Pro	openrouter: baidu/fp8
8	Gemini 3.6 Flash	openrouter: google-vertex/global/flex
10	Kimi K3	kimi_direct: kimi-code-direct
18	Tencent HY 3	openrouter: baidu/fp8
4	Claude Fable 5	openrouter: amazon-bedrock
5	Claude Opus 5	openrouter: amazon-bedrock/claude-on-aws
3	GPT-5.6 Luna Pro	openrouter: openai/flex
2	GPT-5.6 Terra Pro	openrouter: openai/flex
11	Qwen 3.8 Max	openrouter: alibaba
20	Command A Reasoning	cohere_direct: cohere-direct
7	Gemini 3.1 Pro	openrouter: google-vertex/global/flex
15	MiniMax M3	openrouter: together
12	GLM 5.2	openrouter: coreweave/fp4
19	Command A Plus	cohere_direct: cohere-direct
14	DeepSeek V4 Flash	openrouter: cloudflare/fp8
9	Grok 4.5	openrouter: xai/zdr
17	Mistral Large 2512	openrouter: mistral
16	Nemotron 3 Ultra	openrouter: deepinfra/fp4

Table 6: All 20 model routes in the released leaderboard. Routes are selected before generation; automatic fallback is disabled.

B Protocol details

All arms request temperature 0, top-p 1, and seed 20260810 where the provider supports it. The final response budget is 2,048 tokens. Model + Epicure has one call round, one call total, a 4,096-byte result bound, and a 4,096-byte cumulative result bound. At most two provider attempts are allowed, and an ambiguously delivered request is never replayed. The final marker is parsed from plain text for transport portability; direct providers may use native structured generation internally and normalize to the same marker.

C Data fields

Each observation binds model ID, task ID, condition, source status, source and response hashes, returned model and provider, finish reason, response time, final answer, observed and expected choice, normal-parse flag, correctness, and a projected Epicure trace. Each trace event contains its name, structured arguments, error flag, response time, and result hash. The release does not require raw provider credentials or hidden service state.

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